

5.23 Solution: (a) The cylinder can be viewed as a continuous stack of current loops with radius a and an azimuthal current $M\hat{\phi}$, by Eq. (5.103), parallel to the surface. Then, a current loop at z will induce an image current loop with the same radius and same current at $-z$, following the result from Problem 5.17. By Problem 5.18 (a), we know that the force between two current loops that are coaxial and parallel to each other, separated by a distance d , with the same current I in the same direction, is

$$F(d) = \mu_0 I^2 a \frac{\partial}{\partial d} \left[\left(\frac{2}{k(d)} - k(d) \right) K[k(d)] - \frac{2}{k(d)} E[k(d)] \right],$$

where

$$k(d) = \frac{2a}{\sqrt{4a^2 + d^2}}.$$

Then, the force between the cylinder and the infinitely permeable plane surface can be determined by the force between the cylinder and its image,

$$\begin{aligned} F &= \int_0^L dz_1 \int_0^L dz_2 F(z_1 + z_2) \\ &= \mu_0 I^2 a \int_0^L \left\{ \left[\left(\frac{2}{k(z_1 + L)} - k(z_1 + L) \right) K[k(z_1 + L)] - \frac{2}{k(z_1 + L)} E[k(z_1 + L)] \right] \right. \\ &\quad \left. - \left[\left(\frac{2}{k(z_1)} - k(z_1) \right) K[k(z_1)] - \frac{2}{k(z_1)} E[k(z_1)] \right] \right\} dz_1. \end{aligned}$$

The final form of the result suggest that we could try the following calculation,

$$\begin{aligned} &\frac{d}{dz} \left[\frac{z}{k(z)} \left(K[k(z)] - E[k(z)] \right) \right] \\ &= z \frac{d}{dk} \left[\frac{1}{k} \left(K(k) - E(k) \right) \right] \frac{dk}{dz} + \frac{K(k) - E(k)}{k} \\ &= \left[-\frac{z}{k^2} \left(K(k) - E(k) \right) + \frac{z}{k} \left(\frac{E(k)}{k(1-k^2)} - \frac{K(k)}{k} - \frac{E(k)}{k} + \frac{K(k)}{k} \right) \right] \left(-\frac{k(1-k^2)}{z} \right) \\ &\quad + \frac{K(k) - E(k)}{k} \\ &= \frac{2-k^2}{k} K(k) - \frac{2}{k} E(k). \end{aligned}$$

Then, the integral can be performed as

$$\begin{aligned} F &= \mu_0 I^2 a \left[\frac{K[k(z_1 + L)] - E[k(z_1 + L)]}{k(z_1 + L)} \Big|_{z_1=0}^L - \frac{K[k(z_1)] - E[k(z_1)]}{k(z_1)} \Big|_{z_1=0}^L \right] \\ &= 2\mu_0 a L I^2 \left[\frac{K[k(2L)] - E[k(2L)]}{k(2L)} - \frac{K[k(L)] - E[k(L)]}{k(L)} \right]. \end{aligned}$$

Let

$$k_1 = k(2L) = \frac{a}{\sqrt{a^2 + L^2}}, \quad k = k(L) = \frac{2a}{\sqrt{4a^2 + L^2}},$$

and replace I with M , we will finally get

$$F = -2\mu_0 a L M^2 \left[\frac{K(k) - E(k)}{k} - \frac{K(k_1) - E(k_1)}{k_1} \right],$$

which differs Jackson by a minus sign, as the force is attractive.

(b) For $L \gg a$, $k, k_1 \approx 0$. Using the MacLaurin expansion of $K(k)$ and $E(k)$,

$$K(k) = \frac{\pi}{2} \left(1 + \frac{k^2}{4} + \cdots \right), \quad E(k) = \frac{\pi}{2} \left(1 - \frac{k^2}{4} + \cdots \right),$$

we have

$$F \approx -2\mu_0 a L M^2 \left(\frac{\pi}{4} k - \frac{\pi}{4} k_1 \right) \approx -2\mu_0 a L M^2 \cdot \frac{\pi}{4} \cdot \frac{a}{L} = -\frac{\mu_0}{2} \pi a^2 M^2,$$

which agrees with Problem 5.22.